



Python Programming Foundations

Week 1 - First Programs and Basic Values
Day 2: "Move from Input to Output"

1

From Output To Value Flow

Yesterday:

- ▶ we made output visible.

Today

- ▶ We make values move through a program and produce a result.



2



From Simple Output to Numeric Value Flow

We build on what you already know.

Day 1
Simple Output

Code
`print("Hello")`

↓

Output
Hello

→

Day 2
Numeric Value Flow

Code
`total = price * quantity`

↓

Output
Total: 13.50

Now output can come from a calculation.

Output to Value Flow

3

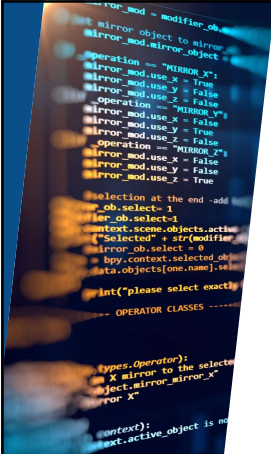
Quick Review

A variable name refers to a stored value.

`'print()'` lets us see that value on the screen.



4



What We Will Use Today

- ▶ Numbers
- ▶ simple expressions
- ▶ result variables
- ▶ `'print()'`

`'input()'` is optional and only useful if it does not distract you from value flow.

5



What We Will "Skip" For Now

- ▶ loops
- ▶ functions
- ▶ lists or dictionaries
- ▶ file saving
- ▶ AI-generated code
- ▶ advanced numeric types



6

Southwest Tech

Today's Beginner Python Working Set

Master a few simple tools first.

Today (Included)
These are in your toolbox.

- print()**
Make Python produce visible output.
- strings**
Work with text and messages.
- variables**
Store and use information in memory.
- run/run**
Run your code. Set results. Try again.

Later (Coming Next)
More tools for bigger jobs.

- input()
- calculations
- loops
- functions
- AI-generated code

★ Focus today. Build confidence. The rest comes next.

Working Set

7

Southwest Tech

Numbers Are Values Too

Python can store numbers just like it stores text.

Numbers can also be used in calculations.

Strings and Numbers as Stored Values

Different kinds of values. Same idea.

String Value
Text is stored as a string.

variable name → stored value

student_name → "Jordan"

⚠ This is text. We use quotes.

Number Value
Numbers are stored as numbers.

variable name → stored value

items_purchased → 3

⚠ This is a number. No quotes.

Both can be stored in variables. Python can calculate with numbers.

8

Southwest Tech

Expressions Create New Values

Combine values. Get a new value.

Expressions Create New Values

Combine values. Get a new value.

Input Values

items_purchased = 3
price_each = 4.50

→

Expression

items_purchased * price_each

→

New Value (Result)

subtotal = 13.50

Example:
▶ 'items_purchased * price_each' creates a subtotal.

⚡ An expression uses values to create a result.

9

Southwest Tech

Result Variables

fx

- ▶ A result variable stores the answer from an expression.
- ▶ Clear names help you explain what the answer means.

10

Southwest Tech

Output Should Mean Something

Printing a number is better when the output tells us what the number means.

'print("Subtotal:", subtotal)' is clearer than printing 'subtotal' alone.

11

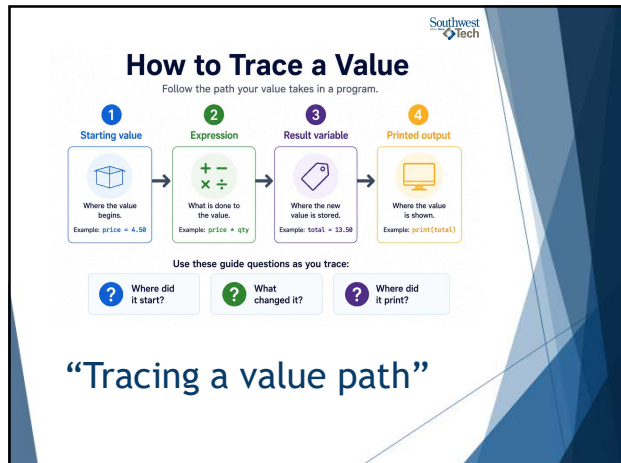
Southwest Tech

Trace The Value

To understand a program, follow one value from start to finish.

Where did it start, what changed it, and where did it print?

12



13

Southwest Tech

The Day 2 Value Path

Starting values become expression results.

Expression results become printed output.

This is also a foundation of knowing how to “debug” a programming issue.

14

Southwest Tech

Demo #1

“Numeric Calculation”

15

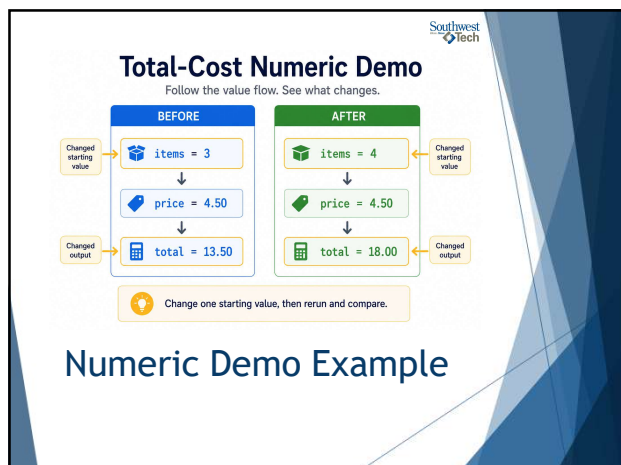
Southwest Tech

What to watch for

As the demo runs, watch for two things:

- ▶ which values are stored
- ▶ which lines make output appear

16



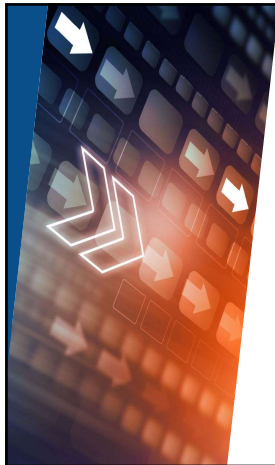
17

Southwest Tech

Optional Demo #2

“Simple Input”

18

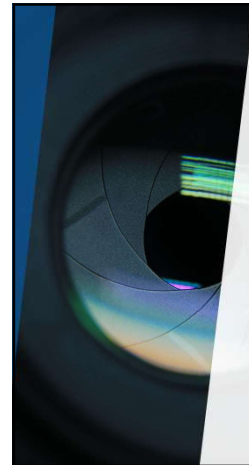


Southwest
Tech

Optional Demo: Simple ``input()``

- ▶ ``input()`` can collect a user response.
- ▶ That response still becomes a value the program must trace.

19



Optional Demo #3

"Numeric Conversion"

20



Southwest
Tech

Optional Demo: Numeric Input Needs Conversion

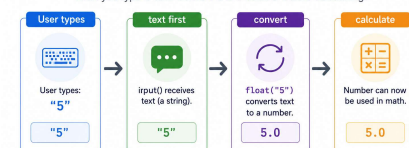
- ▶ ``input()`` gives Python text.
- ▶ *For math, numeric input must be converted first.*

21

Southwest
Tech

Numeric Input: Convert Before Math

What you type is text. Convert it to a number before calculating.



Key idea: Your keyboard input is text first. Convert it to a number, then do the math.

You may use ``input()`` if needed. It is optional for this assignment.

Numeric Conversion Example

22



Common Failure: Losing The Value

- ▶ When several variables appear ... it is easy to lose the path.

Slow down and ask:

- ▶ "Where did this value come from?"

23

Southwest
Tech

Common Failure: ``input()`` Can Distract



- ▶ ``input()`` is useful, but it is not the main goal today.
- ▶ The main goal is understanding how a value becomes output.

24



Southwest
Tech

From Demo to Lab

In Lab/Assignment:

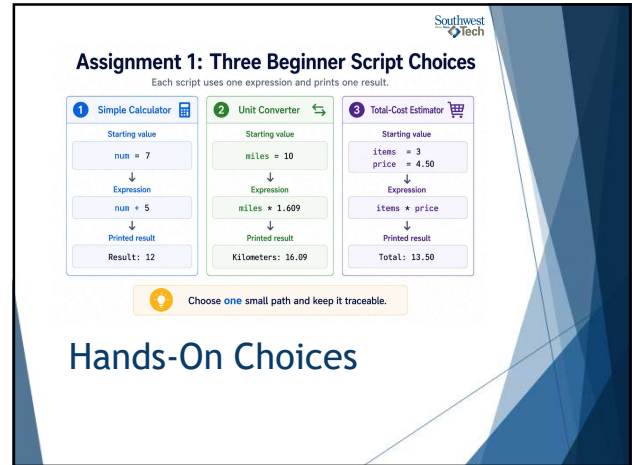
- ▶ Build one small script that starts with numbers and prints a result.

Good choices:

- ▶ Calculator
- ▶ Converter
- ▶ Total-cost estimator

This is a continuation of the first assignment.

25



Southwest
Tech

Assignment 1: Three Beginner Script Choices

Each script uses one expression and prints one result.

1 Simple Calculator

Starting value

num = 7

↓

Expression

num + 5

↓

Printed result

Result: 12

2 Unit Converter

Starting value

miles = 10

↓

Expression

miles * 1.609

↓

Printed result

Kilometers: 16.09

3 Total-Cost Estimator

Starting value

items = 3
price = 4.50

↓

Expression

items * price

↓

Printed result

Total: 13.50

Choose one small path and keep it traceable.

Hands-On Choices

26

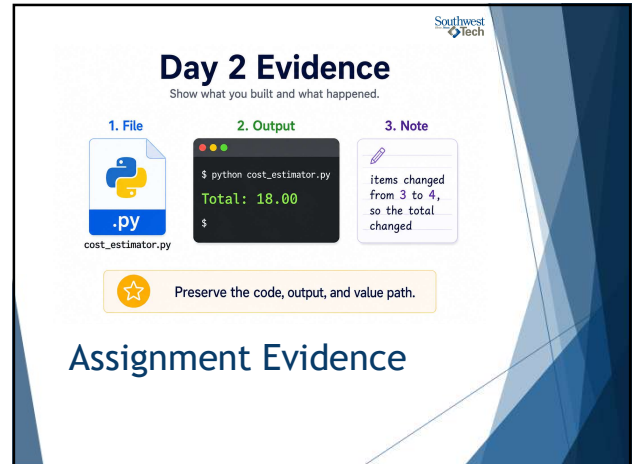


Southwest
Tech

Assignment Evidence

- ▶ Preserve the '.py' file and one example of output.
- ▶ Be ready to explain the value path from starting number to printed result.

27



Southwest
Tech

Day 2 Evidence

Show what you built and what happened.

1. File

 .py

cost_estimator.py

2. Output

```
$ python cost_estimator.py
Total: 18.00
$
```

3. Note

items changed from 3 to 4, so the total changed.

Preserve the code, output, and value path.

Assignment Evidence

28



Southwest
Tech

Closing Success Check

By the end of today you should be able to:

- ▶ store numbers,
- ▶ calculate a result
- ▶ print it clearly.
- ▶ You should also be able to trace where the printed result came from.

29



Southwest
Tech

Thank you!

Have Fun!

30